

Sienna WAV frame — MANUFACTURING PACKAGE — index (rev A, 2026-10-03)

One folder per part: <PART>_drawing.pdf + DXF files. Units: inch. Quantities are per vehicle (FR = right side, FL = left side mirror).

Item	Description	Material	Flat / length (in)	Bends	Qty/veh	Process
FR-1	Plate with ears — flat pattern (2 bends 90°)	10 ga 0.135"	8.512 × 13.851	2	2	laser + brake
FR-2	FR-2 one piece (top + 2 U legs)	10 ga 0.135"	22.078 × 3.250	6	2	laser + brake
FR-3	Channel — flat pattern (2 bends 90°)	11 ga 0.120"	4.084 × 7.734	2	2	laser + brake
FR-4-1	Wrap strip (ear, top, front, bottom, slant)	10 ga 0.135"	18.746 × 2.716	9	2	laser + brake
FR-4-2	Side plate A (2 holes Ø0.512)	10 ga 0.135"	6.500 × 6.875	0	2	laser
FR-4-3	Side plate B (holes Ø0.512 + Ø0.34)	10 ga 0.135"	6.500 × 6.875	0	2	laser
FR-4 tube	Tube OD 1.5 / ID 1.181 — cut 3.689" (through bottom and to	DOM tube OD 1.500" × ID 1.181"	L 3.689	0	2	saw
FR-5	U-body (2 sides + plate A-C)	10 ga 0.135"	10.643 × 6.198	2	2	laser + brake
FR-6 tube	Tube 3×3 — cut 14 ¼"	Sq. tube 3" × 3" × 3/16" wall	L 14.250	0	2	saw
FR-7	FR-7 one piece (back + bottom + 2 sides)	10 ga 0.135"	9.801 × 8.160	3	2	laser + brake
FR-8	Channel (top + 2 sides)	11 ga 0.120"	6.209 × 12.000	2	2	laser + brake
FR-9 tube	Tube 3×3 — cut 8"	Sq. tube 3" × 3" × 3/16" wall	L 8.000	0	2	saw
FR-10-1	Part 1 "U-body" — 2 sides + wall A-H + flange E-F (3 bends	10 ga 0.135"	12.287 × 11.214	3	2	laser + brake
FR-10-2	Part 2 "strip" — A-B · B-C · shelf · inner plate (3 bends)	10 ga 0.135"	12.478 × 2.716	3	2	laser + brake
FR-10 tube	Tube OD 1.5 / ID 1.181 — cut 3.42" (through A-B and the sh	DOM tube OD 1.500" × ID 1.181"	L 3.420	0	2	saw
FR-11	FR-11 plate (flat)	11 ga 0.120"	5.125 × 2.000	0	4	laser
FF-1	Channel with ears — flat pattern, one piece (4 bends 90°,	10 ga 0.135"	20.139 × 6.818	4	1	laser + brake
FF-3-1	FF3-1 top plate — flat pattern, 10 bends 90° down	11 ga 0.120"	39.403 × 15.073	10	1	laser + brake
FF-3-2	FF3-2 lower plate — flat pattern, 2 bends 90° up	11 ga 0.120"	39.000 × 14.595	2	1	laser + brake
FF-3-3	Passage wall 1⅞ × 4.69" (instead of the tube)	11 ga 0.120"	4.687 × 1.125	0	2	laser
FF-3-4	Passage wall 1⅞ × 5.01" (instead of the tube)	11 ga 0.120"	5.013 × 1.125	0	2	laser
FF-4 tube	Square tube 1 ½" — cut 39"	Sq. tube 1 ½" × 1 ½" × 3/16" wall	L 39.000	0	1	saw

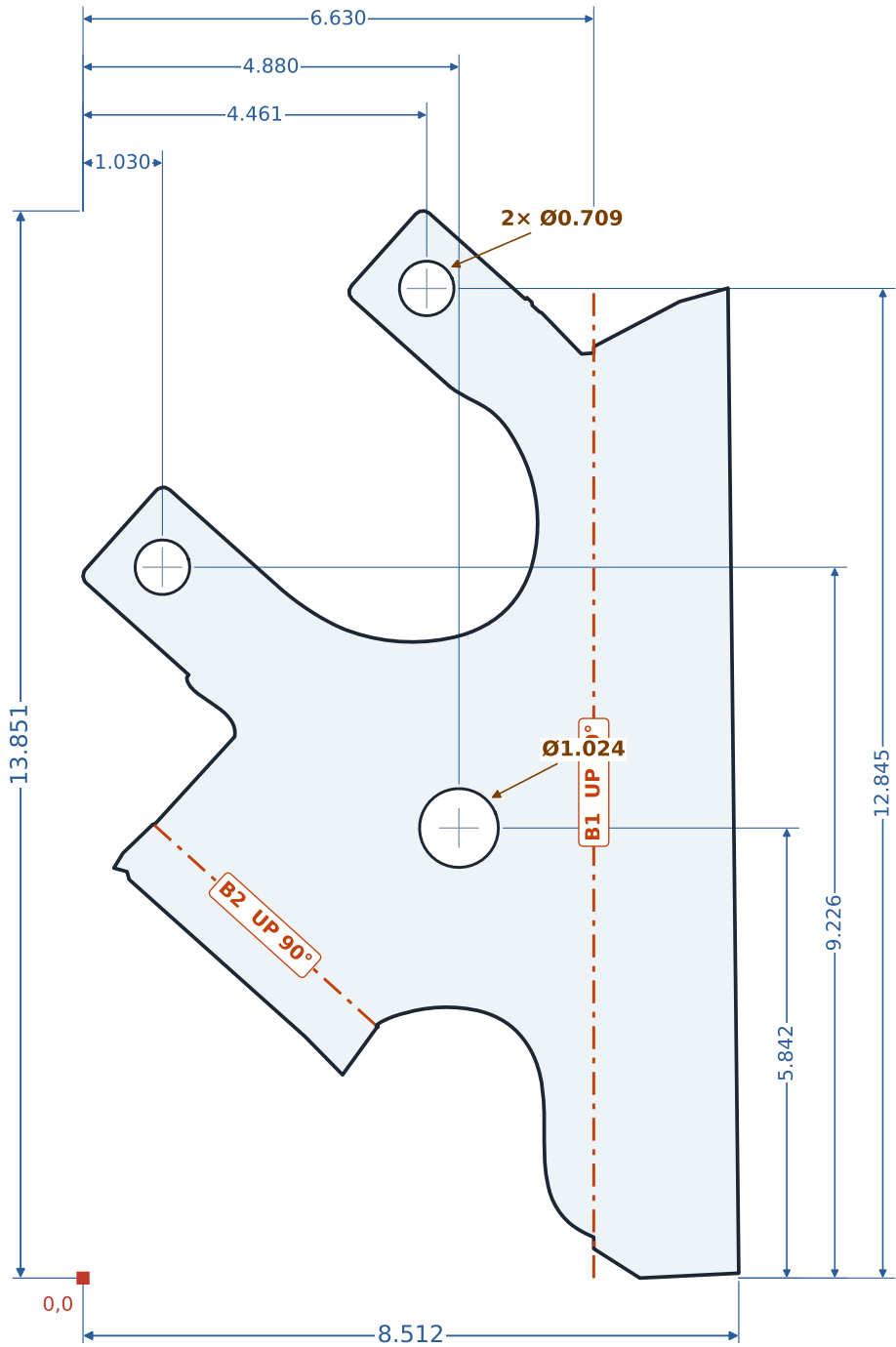
MATERIAL PER VEHICLE (net, without nesting scrap)

- 10 ga 0.135" (3.42 mm): 907 in² net (6.30 ft²), ≈ 34.6 lb
- 11 ga 0.120" (3.04 mm): 1214 in² net (8.43 ft²), ≈ 41.2 lb
- DOM round steel tube OD 1.500" × ID 1.181" (wall ≈ 0.160"): 14.22 in total length
- Square steel tube 3" × 3" × 3/16" wall (A500): 44.50 in total length
- Square steel tube 1 ½" × 1 ½" × 3/16" wall (A500): 39.00 in total length

FR-1 · Plate with ears — flat pattern (2 bends 90°)

10 ga 0.135" (3.42 mm) · HR steel sheet A1011 CS · flat blank 8.512" × 13.851" · ≈ 2.28 lb each · QTY 1 FR + 1 FL = 2 per vehicle

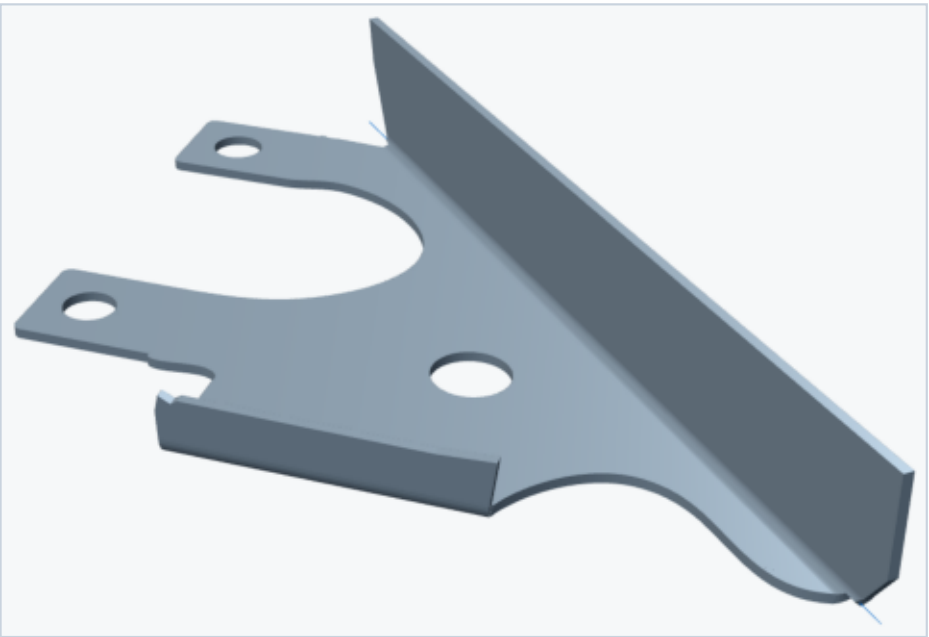
FLAT PATTERN (as laser-cut, marked side up)



NOTES

- 1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - 2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - 3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - 4. Deburr all edges, remove dross.
 - 5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.
- * bend 90° up on dashed lines

FORMED PART (after bending)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	UP (toward)	R=t 0.135	1	X = 6.630
B2	90.0°	UP (toward)	R=t 0.135	2	(0.922, 5.882)→(3.803, 3.278)

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø1.024	4.880	5.842	laser
H2	Ø0.709	4.461	12.845	laser
H3	Ø0.709	1.030	9.226	laser

FILES

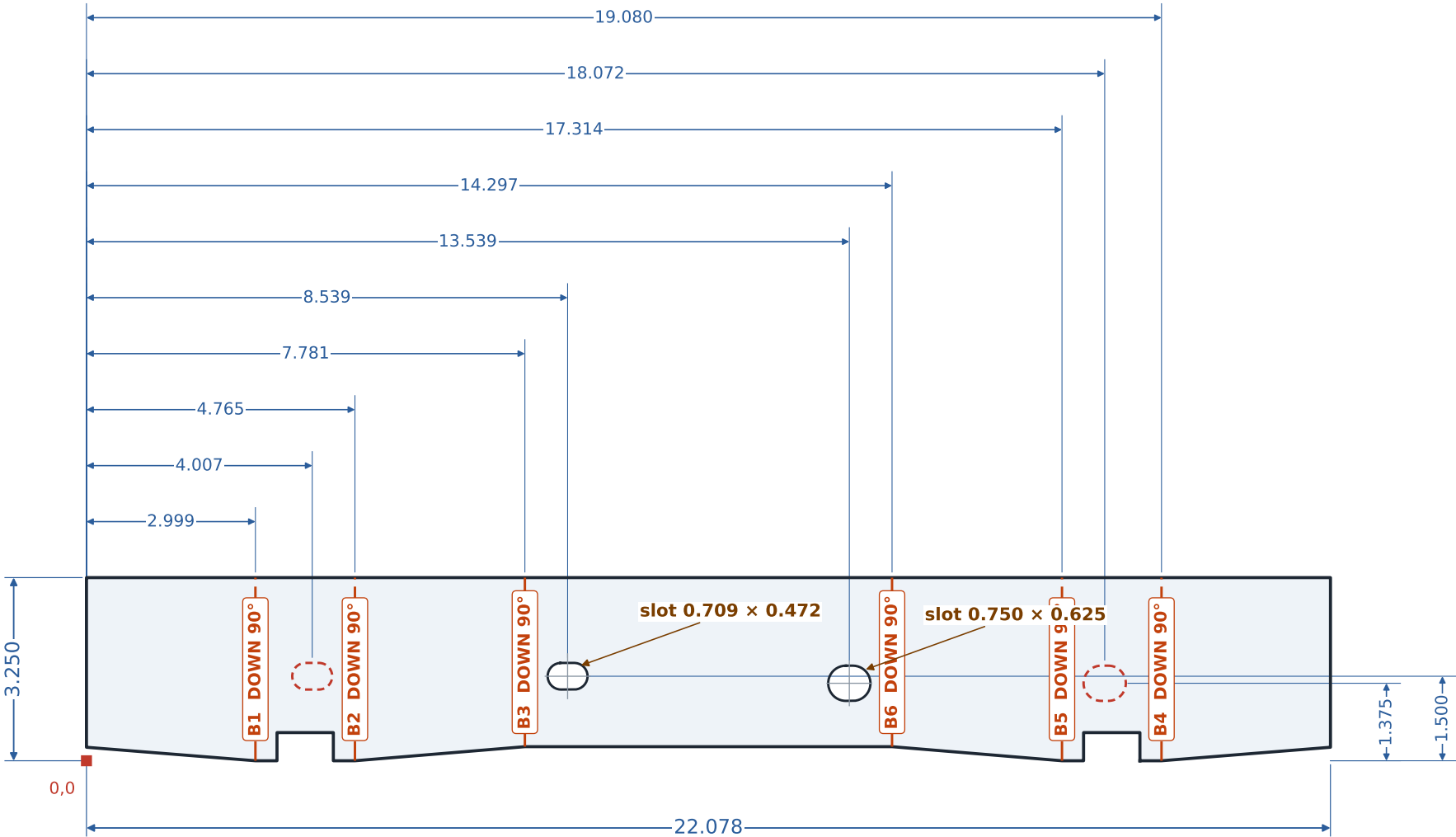
- FR-1_FR_cut.dxf (right, qty 1 / vehicle)
- FR-1_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-1_ref.dxf (reference: + bend lines, drill-later)

FR-1	Side plate with ears and 2 flanges	HR steel sheet A1011 CS 10 ga 0.135" (3.42 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FR-1	Plate with ears — flat pattern (2 bends 90°)	QTY 1 FR + 1 FL = 2 per vehicle	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 1/1

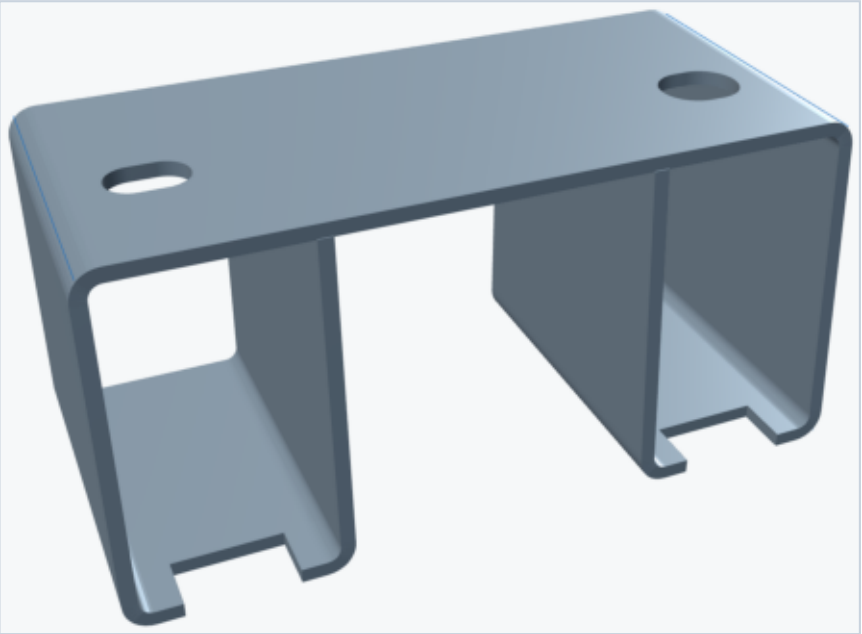
FR-2 · FR-2 one piece (top + 2 U legs)

10 ga 0.135” (3.42 mm) · HR steel sheet A1011 CS · flat blank 22.078” × 3.250” · ≈ 2.55 lb each · QTY 1 FR + 1 FL = 2 per vehicle

FLAT PATTERN (as laser-cut, marked side up)



FORMED PART (after bending)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	DOWN (away)	R=t 0.135	1	X = 2.999
B2	90.0°	DOWN (away)	R=t 0.135	3	X = 4.765
B3	90.0°	DOWN (away)	R=t 0.135	5	X = 7.781
B4	90.0°	DOWN (away)	R=t 0.135	2	X = 19.080
B5	90.0°	DOWN (away)	R=t 0.135	4	X = 17.314
B6	90.0°	DOWN (away)	R=t 0.135	6	X = 14.297

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	slot 0.709×0.472	8.539	1.500	laser
H2	slot 0.750×0.625	13.539	1.375	laser
D1	slot 0.709×0.472	4.007	1.500	DRILL LATER
D2	slot 0.750×0.625	18.072	1.375	DRILL LATER

FILES

- FR-2_FR_cut.dxf (right, qty 1 / vehicle)
- FR-2_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-2_ref.dxf (reference: + bend lines, drill-later)

NOTES

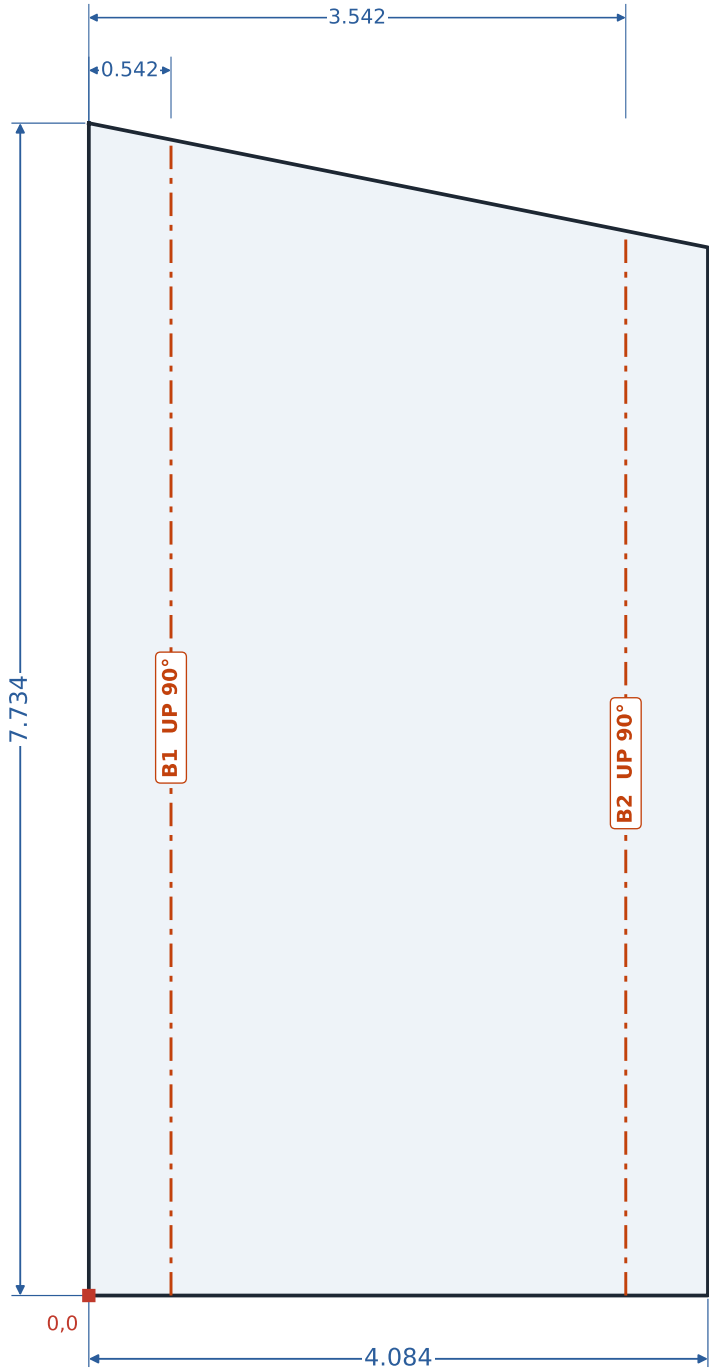
- Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - Deburr all edges, remove dross.
 - Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.
- * one blank, 6 bends 90° the same way (R = t): inner walls first, then the bottoms, then the outer walls; top ovals laser cut; do NOT cut the leg-bottom ovals — drill after welding through the top ovals; ½" notch at the front edge of each bottom

FR-2 Bridge bracket (top + 2 U legs)	HR steel sheet A1011 CS 10 ga 0.135” (3.42 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FR-2 · FR-2 one piece (top + 2 U legs)	QTY 1 FR + 1 FL = 2 per vehicle	Tol.: cut ±0.010” · bend ±1° · IR = t	2026-10-03	SHEET 1/1

FR-3 · Channel — flat pattern (2 bends 90°)

11 ga 0.120" (3.04 mm) · HR steel sheet A1011 CS · flat blank 4.084" × 7.734" · ≈ 1.01 lb each · QTY 1 FR + 1 FL = 2 per vehicle

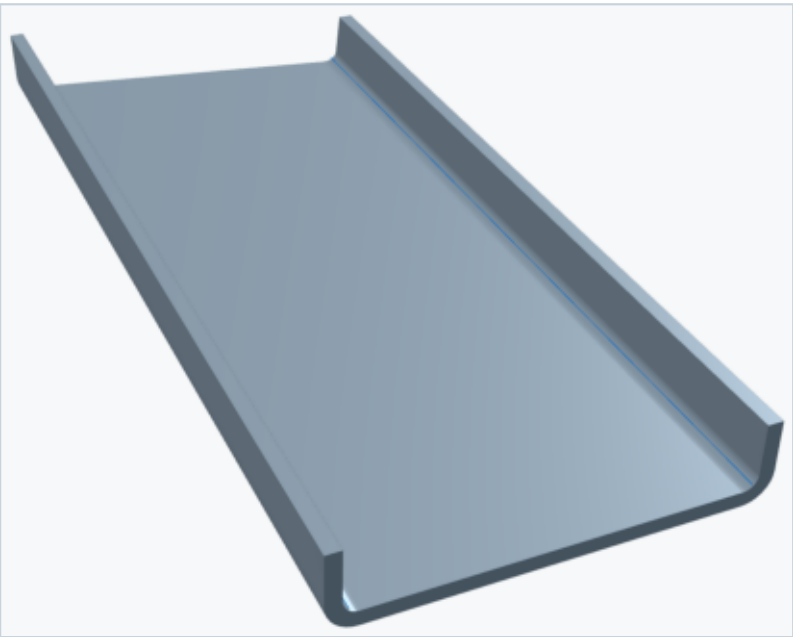
FLAT PATTERN (as laser-cut, marked side up)



NOTES

- 1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - 2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - 3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - 4. Deburr all edges, remove dross.
 - 5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.
- * bend 90° on dashed lines, 3/4 flanges inward

FORMED PART (after bending)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	UP (toward)	R=t 0.120	1	X = 0.542
B2	90.0°	UP (toward)	R=t 0.120	2	X = 3.542

FILES

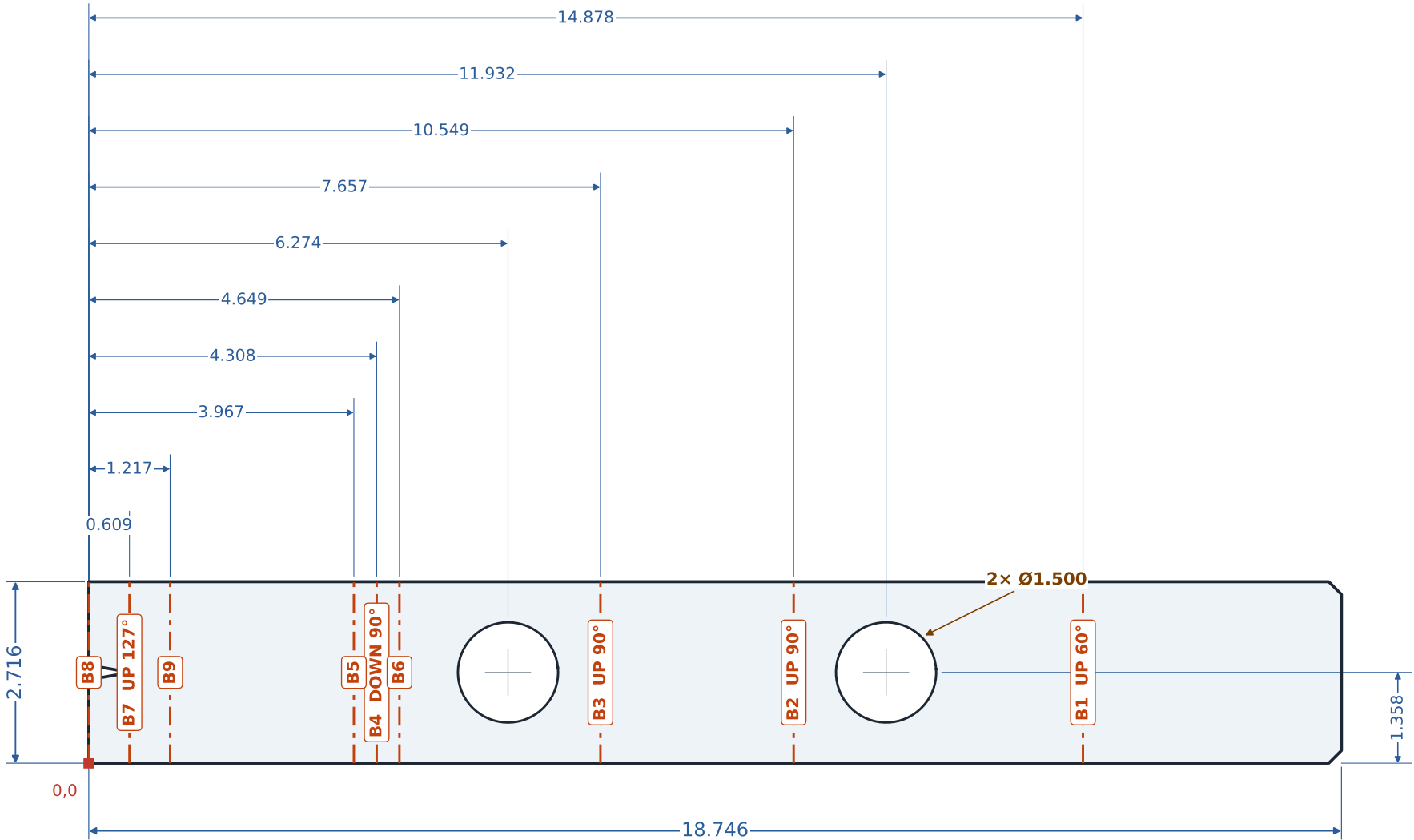
- FR-3_FR_cut.dxf (right, qty 1 / vehicle)
- FR-3_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-3_ref.dxf (reference: + bend lines, drill-later)

FR-3Channel	HR steel sheet A1011 CS 11 ga 0.120" (3.04 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FR-3 · Channel — flat pattern (2 bends 90°)	QTY 1 FR + 1 FL = 2 per vehicle	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 1/1

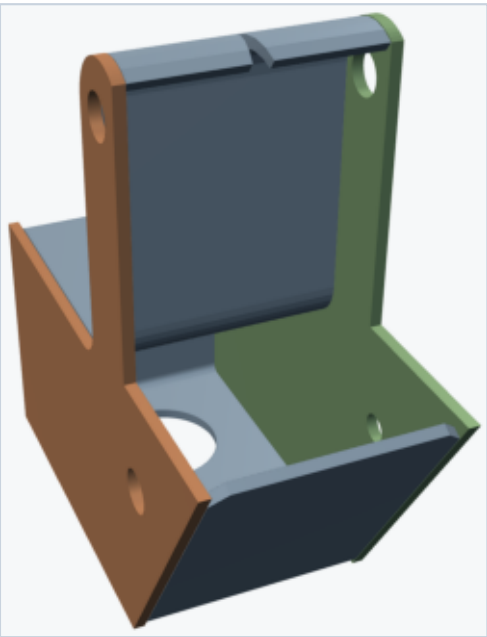
FR-4-1 · Wrap strip (ear, top, front, bottom, slant)

10 ga 0.135" (3.42 mm) · HR steel sheet A1011 CS · flat blank 18.746" × 2.716" · ≈ 1.80 lb each · QTY 1 FR + 1 FL = 2 per vehicle

FLAT PATTERN (as laser-cut, marked side up)



FORMED PART (after bending)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	60.0°	UP (toward)	R=t 0.135	5	X = 14.878
B2	90.0°	UP (toward)	R=t 0.135	4	X = 10.549
B3	90.0°	UP (toward)	R=t 0.135	3	X = 7.657
B4	90.0°	DOWN (away)	R 0.380	2	X = 4.308
B5	—	radius zone edge	—		X = 3.967
B6	—	radius zone edge	—		X = 4.649
B7	126.9°	UP (toward)	R 0.490	1	X = 0.609
B8	—	radius zone edge	—		X = 0.000
B9	—	radius zone edge	—		X = 1.217

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø1.500	6.274	1.358	laser
H2	Ø1.500	11.932	1.358	laser

FILES

- FR-4-1_FR_cut.dxf (right, qty 1 / vehicle)
- FR-4-1_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-4-1_ref.dxf (reference: + bend lines, drill-later)

NOTES

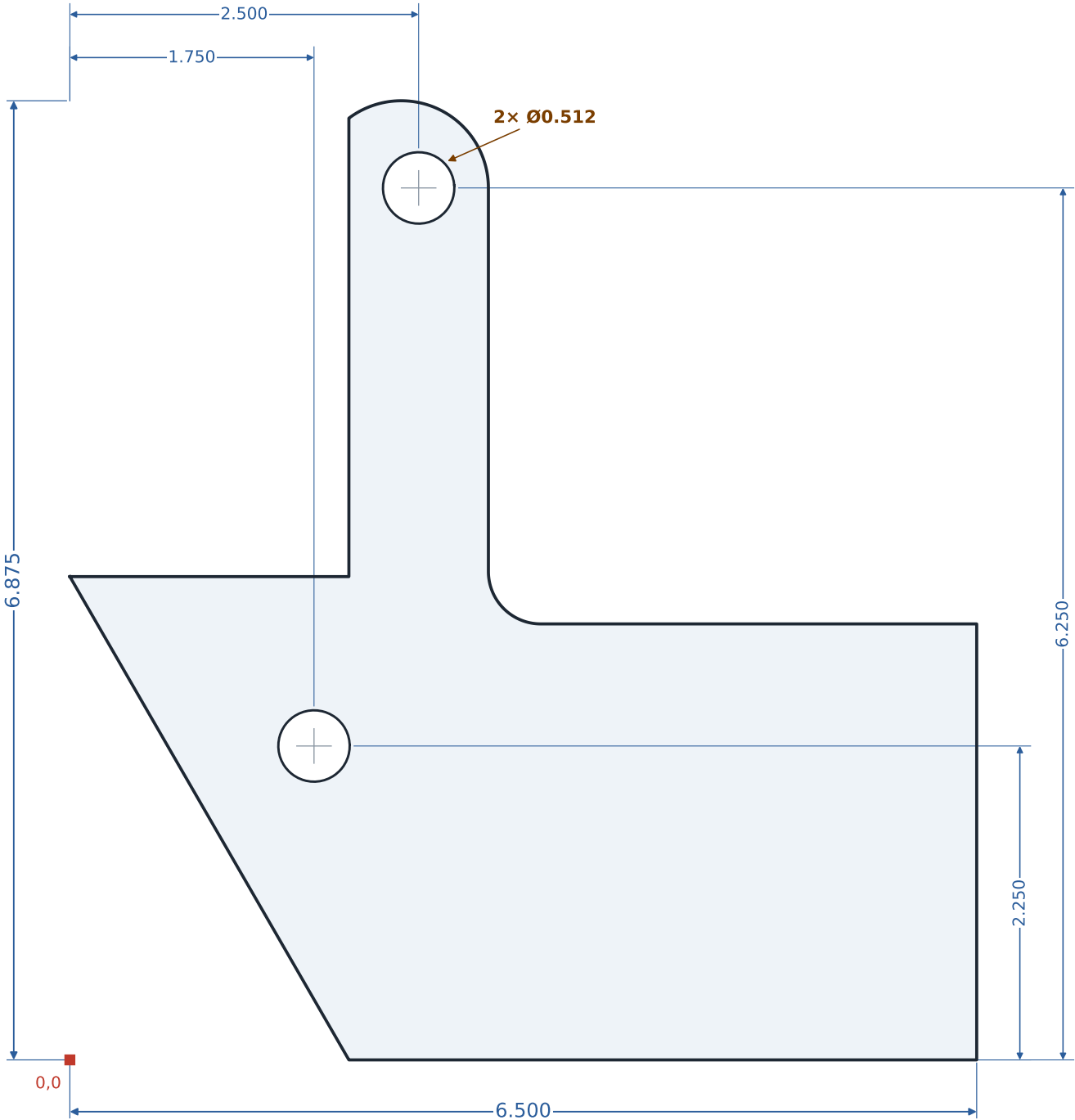
- Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - Deburr all edges, remove dross.
 - Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.
- * arc C-D bent over R 0.48; R ⅜" at E; no corner reliefs

FR-4	Bracket: wrap strip + 2 side plates + tube	HR steel sheet A1011 CS 10 ga 0.135" (3.42 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FR-4-1 · Wrap strip (ear, top, front, bottom, slant)	QTY 1 FR + 1 FL = 2 per vehicle		Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 1/4

FR-4-2 · Side plate A (2 holes Ø0.512)

10 ga 0.135" (3.42 mm) · HR steel sheet A1011 CS · flat blank 6.500" × 6.875" · ≈ 0.79 lb each · QTY 1 FR + 1 FL = 2 per vehicle

FLAT PATTERN (as laser-cut, marked side up)



NOTES

- 1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - 2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - 3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - 4. Deburr all edges, remove dross.
 - 5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.
- * flat, no bends

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø0.512	2.500	6.250	laser
H2	Ø0.512	1.750	2.250	laser

FILES

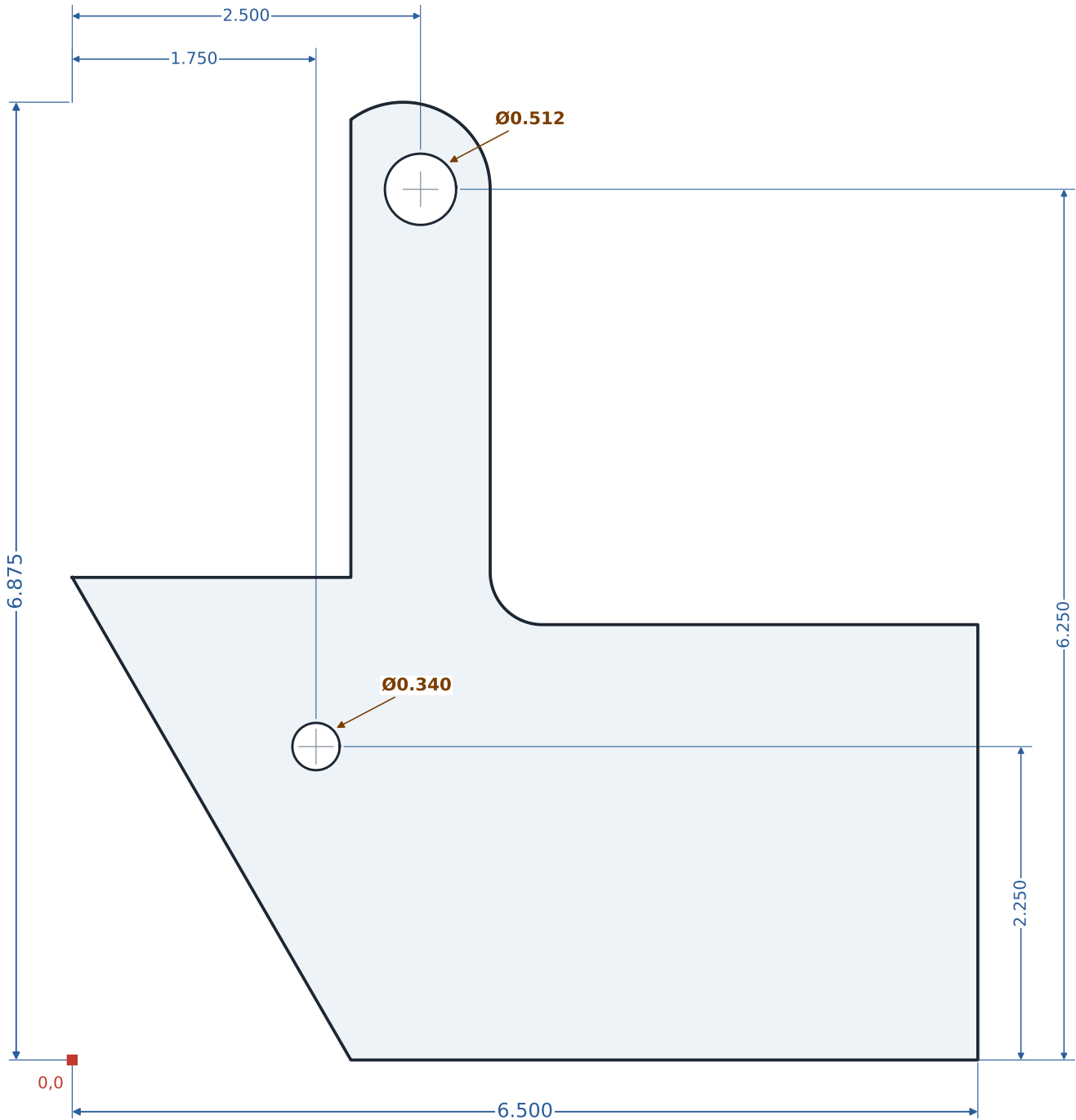
- FR-4-2_FR_cut.dxf (right, qty 1 / vehicle)
- FR-4-2_FL_cut.dxf (left = mirror, qty 1 / vehicle)

FR-4	Bracket: wrap strip + 2 side plates + tube	HR steel sheet A1011 CS 10 ga 0.135" (3.42 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FR-4-2 · Side plate A (2 holes Ø0.512)		QTY 1 FR + 1 FL = 2 per vehicle	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 2/4

FR-4-3 · Side plate B (holes Ø0.512 + Ø0.34)

10 ga 0.135" (3.42 mm) · HR steel sheet A1011 CS · flat blank 6.500" × 6.875" · ≈ 0.80 lb each · QTY 1 FR + 1 FL = 2 per vehicle

FLAT PATTERN (as laser-cut, marked side up)



NOTES

- 1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - 2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - 3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - 4. Deburr all edges, remove dross.
 - 5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.
- * flat, no bends

HOLES (centre from 0,0)

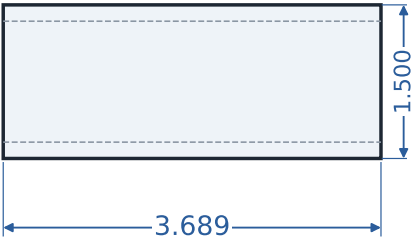
Hole	Size	X	Y	Made by
H1	Ø0.512	2.500	6.250	laser
H2	Ø0.340	1.750	2.250	laser

FILES

- FR-4-3_FR_cut.dxf (right, qty 1 / vehicle)
- FR-4-3_FL_cut.dxf (left = mirror, qty 1 / vehicle)

FR-4	Bracket: wrap strip + 2 side plates + tube	HR steel sheet A1011 CS 10 ga 0.135" (3.42 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FR-4-3 · Side plate B (holes Ø0.512 + Ø0.34)	QTY 1 FR + 1 FL = 2 per vehicle	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 3/4	

FR-4 · tube cut list



Tube OD 1.5 / ID 1.181 — cut 3.689" (through bottom and top)

Stock: DOM round steel tube OD 1.500" × ID 1.181" (wall ≈ 0.160")

Cut length 3.689" ±1/32", square ends, deburr

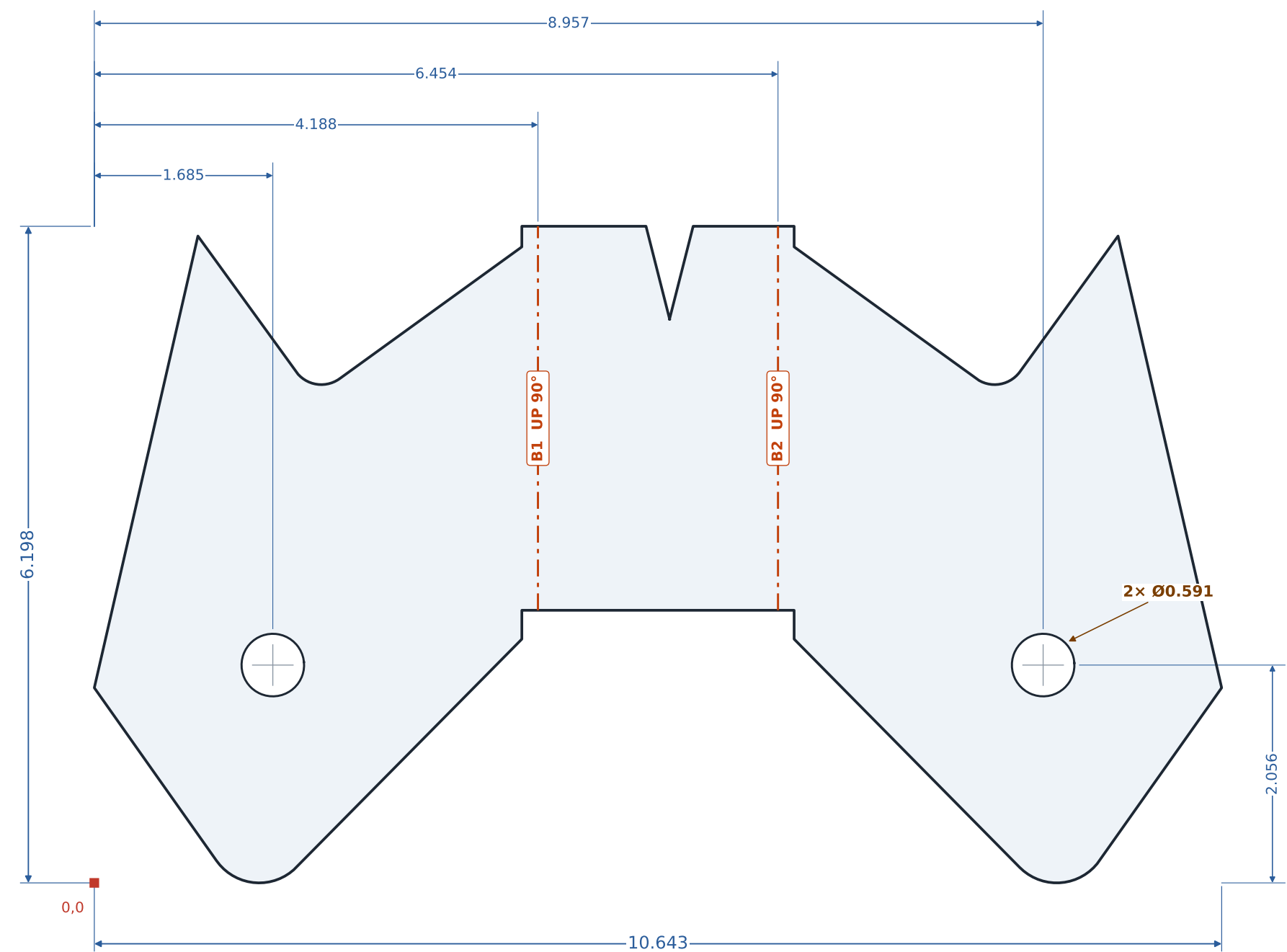
QTY 1 per side × 2 (FR + FL) = 2 per vehicle

FR-4	Bracket: wrap strip + 2 side plates + tube	see stock above	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
Tube cut list		see quantities above	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 4/4

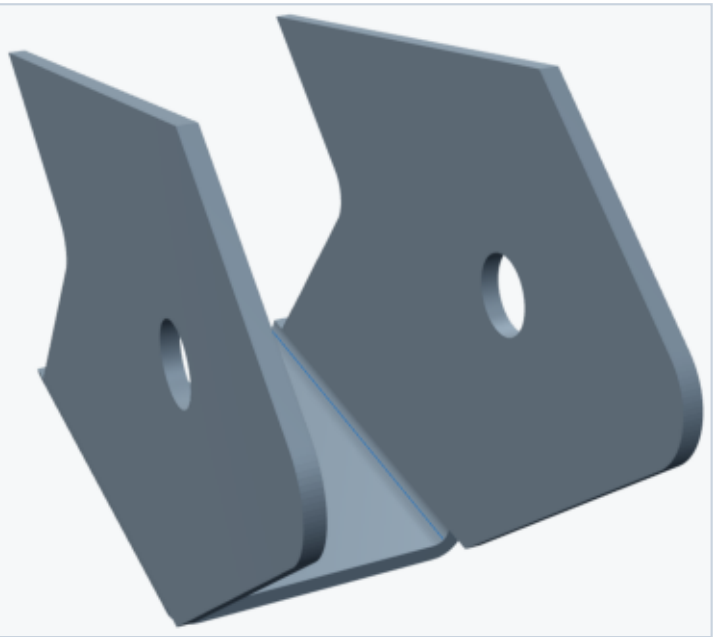
FR-5 · U-body (2 sides + plate A-C)

10 ga 0.135" (3.42 mm) · HR steel sheet A1011 CS · flat blank 10.643" × 6.198" · ≈ 1.57 lb each · QTY 1 FR + 1 FL = 2 per vehicle

FLAT PATTERN (as laser-cut, marked side up)



FORMED PART (after bending)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	UP (toward)	R=t 0.135	1	X = 4.188
B2	90.0°	UP (toward)	R=t 0.135	2	X = 6.454

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø0.591	1.685	2.056	laser
H2	Ø0.591	8.957	2.056	laser

FILES

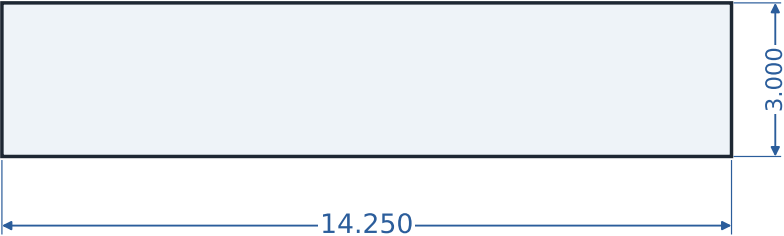
- FR-5_FR_cut.dxf (right, qty 1 / vehicle)
- FR-5_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-5_ref.dxf (reference: + bend lines, drill-later)

NOTES

- Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - Deburr all edges, remove dross.
 - Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.
- * 90° bends, R = t

FR-5U-body bracket	HR steel sheet A1011 CS 10 ga 0.135" (3.42 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FR-5 · U-body (2 sides + plate A-C)	QTY 1 FR + 1 FL = 2 per vehicle	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 1/1

FR-6 · tube cut list



Tube 3×3 — cut 14 ¼"

Stock: Square steel tube 3" × 3" × 3/16" wall (A500)

Cut length 14.250" ±1/32", square ends, deburr

QTY 1 per side × 2 (FR + FL) = 2 per vehicle

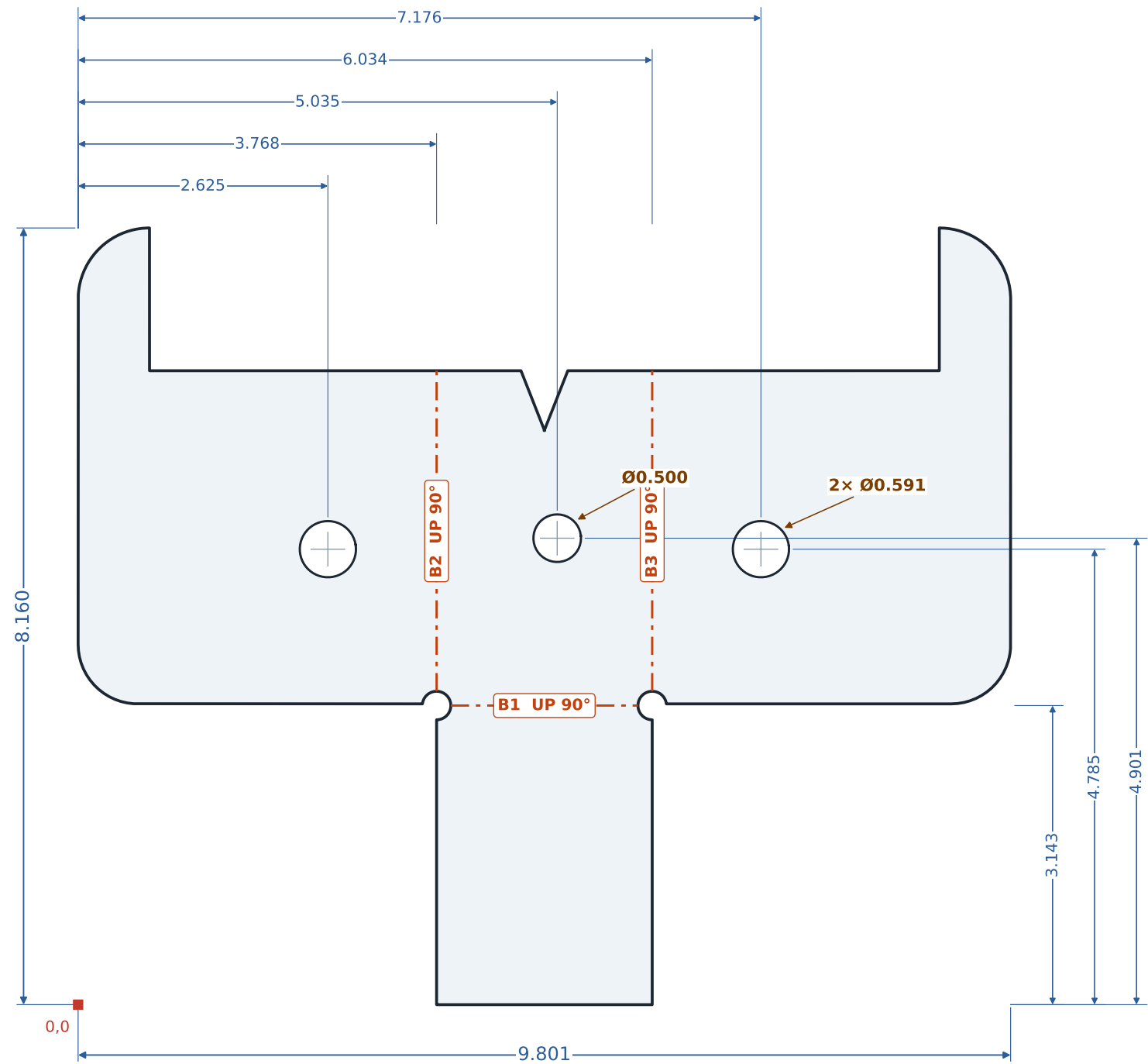
wall 3/16"

FR-6	Frame tube 3×3	see stock above	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
Tube cut list		see quantities above	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 1/1

FR-7 · FR-7 one piece (back + bottom + 2 sides)

10 ga 0.135" (3.42 mm) · HR steel sheet A1011 CS · flat blank 9.801" × 8.160" · ≈ 1.61 lb each · QTY 1 FR + 1 FL = 2 per vehicle

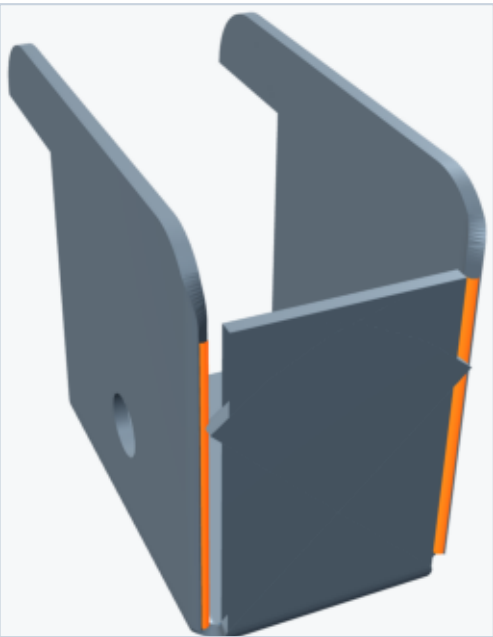
FLAT PATTERN (as laser-cut, marked side up)



NOTES

- 1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - 2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - 3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - 4. Deburr all edges, remove dross.
 - 5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.
- * one blank, 3 bends 90° the same way (R = t): bottom first, then the sides; Ø0.3 round reliefs in the two corners; after bending weld the side-to-bottom seams; V notch and Ø0.5 hole in the back, Ø0.591 holes in the sides

FORMED PART (after bending)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	UP (toward)	R=t 0.135	1	Y = 3.143
B2	90.0°	UP (toward)	R=t 0.135	2	X = 3.768
B3	90.0°	UP (toward)	R=t 0.135	3	X = 6.034

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø0.591	2.625	4.785	laser
H2	Ø0.591	7.176	4.785	laser
H3	Ø0.500	5.035	4.901	laser

FILES

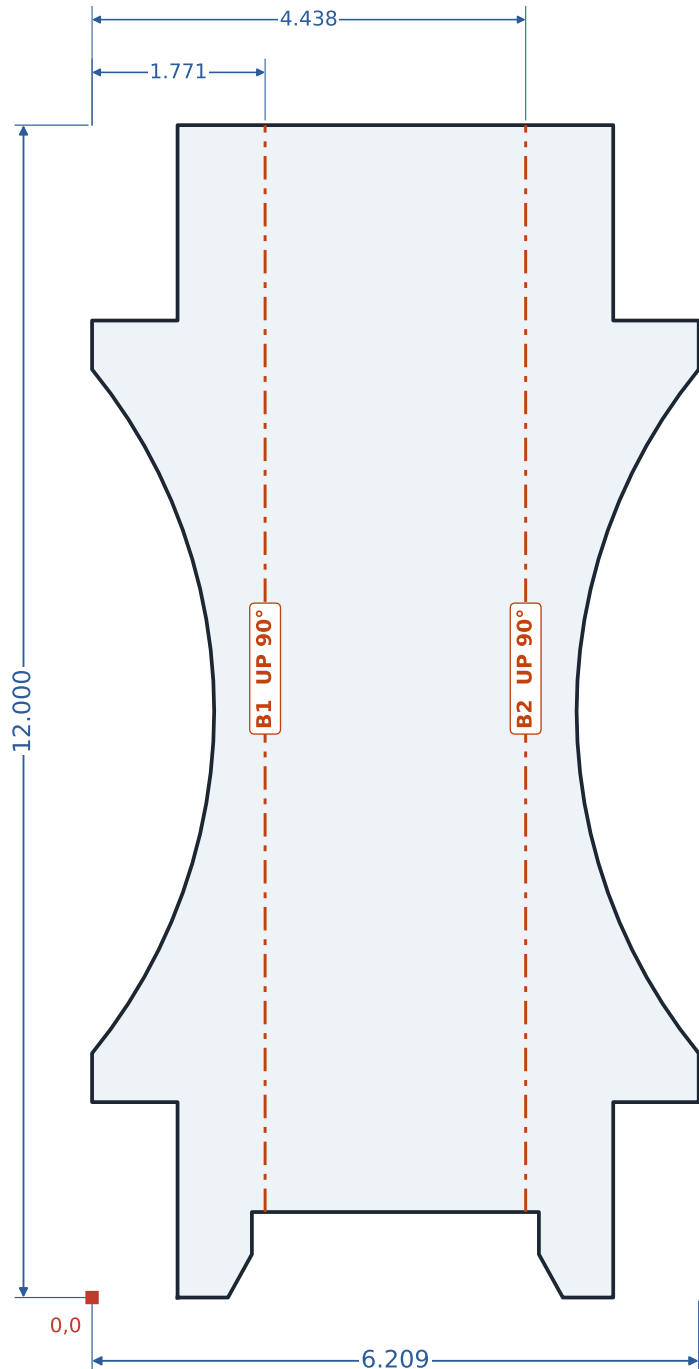
- FR-7_FR_cut.dxf (right, qty 1 / vehicle)
- FR-7_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-7_ref.dxf (reference: + bend lines, drill-later)

FR-7	Tray bracket (back + bottom + 2 sides)	HR steel sheet A1011 CS 10 ga 0.135" (3.42 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FR-7 · FR-7 one piece (back + bottom + 2 sides)		QTY 1 FR + 1 FL = 2 per vehicle	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 1/1

FR-8 · Channel (top + 2 sides)

11 ga 0.120" (3.04 mm) · HR steel sheet A1011 CS · flat blank 6.209" × 12.000" · ≈ 1.79 lb each · QTY 1 FR + 1 FL = 2 per vehicle

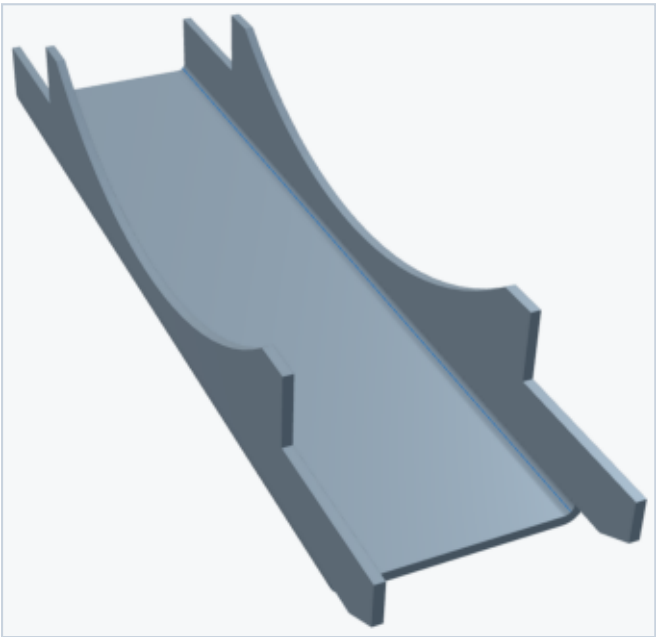
FLAT PATTERN (as laser-cut, marked side up)



NOTES

- 1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - 2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - 3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - 4. Deburr all edges, remove dross.
 - 5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.
- * 90° bends, R = t; relief cut at B

FORMED PART (after bending)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	UP (toward)	R=t 0.120	1	X = 1.771
B2	90.0°	UP (toward)	R=t 0.120	2	X = 4.438

FILES

- FR-8_FR_cut.dxf (right, qty 1 / vehicle)
- FR-8_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-8_ref.dxf (reference: + bend lines, drill-later)

FR-8 Channel (top + 2 sides)	HR steel sheet A1011 CS 11 ga 0.120" (3.04 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FR-8 · Channel (top + 2 sides)	QTY 1 FR + 1 FL = 2 per vehicle	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 1/1

FR-9 · tube cut list



Tube 3×3 — cut 8"

Stock: Square steel tube 3" × 3" × 3/16" wall (A500)

Cut length 8.000" ±1/32", square ends, deburr

QTY 1 per side × 2 (FR + FL) = 2 per vehicle

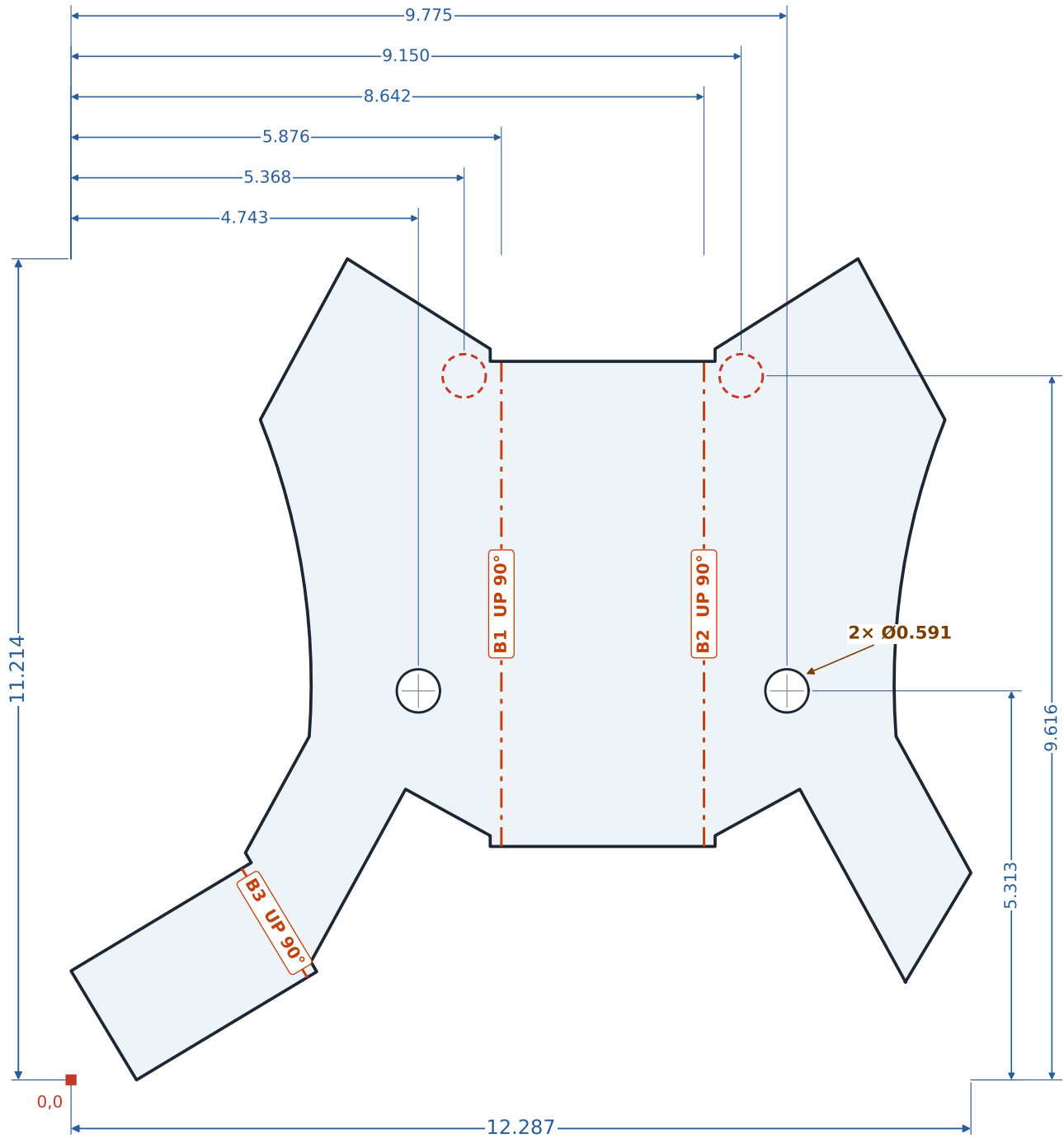
wall 3/16"

FR-9	Frame tube 3×3	see stock above	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
Tube cut list		see quantities above	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 1/1

FR-10-1 · Part 1 "U-body" — 2 sides + wall A-H + flange E-F (3 bends)

10 ga 0.135" (3.42 mm) · HR steel sheet A1011 CS · flat blank 12.287" × 11.214" · ≈ 2.55 lb each · QTY 1 FR + 1 FL = 2 per vehicle

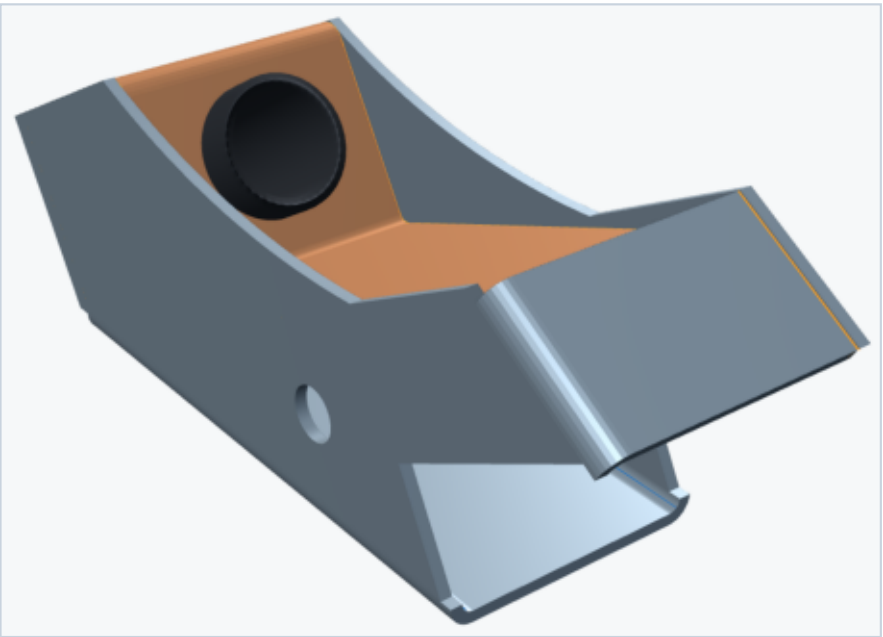
FLAT PATTERN (as laser-cut, marked side up)



NOTES

- 1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - 2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - 3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - 4. Deburr all edges, remove dross.
 - 5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.
- * bends 90° up, R = t; the 2 Ø0.591 holes near A (½" from A-B, ⅝" from A-H) are not in the flat — drill after bending

FORMED PART (after bending)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	UP (toward)	R=t 0.135	2	X = 5.876
B2	90.0°	UP (toward)	R=t 0.135	3	X = 8.642
B3	90.0°	UP (toward)	R=t 0.135	1	(2.331, 2.888)→(3.225, 1.399)

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø0.591	4.743	5.313	laser
H2	Ø0.591	9.775	5.313	laser
D1	Ø0.591	5.368	9.616	DRILL LATER
D2	Ø0.591	9.150	9.616	DRILL LATER

FILES

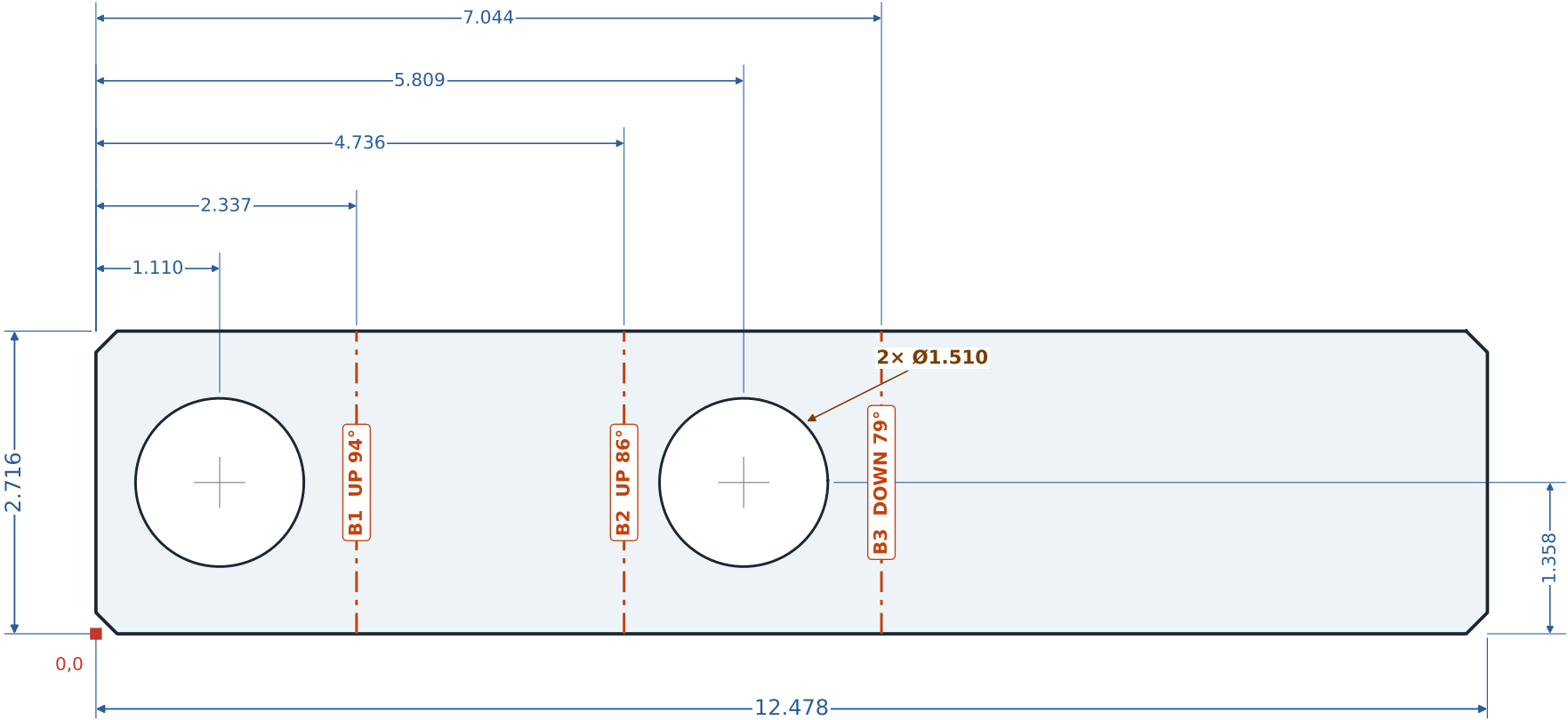
- FR-10-1_FR_cut.dxf (right, qty 1 / vehicle)
- FR-10-1_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-10-1_ref.dxf (reference: + bend lines, drill-later)

FR-10	Bracket: U-body + strip + tube	HR steel sheet A1011 CS 10 ga 0.135" (3.42 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FR-10-1 · Part 1 "U-body" — 2 sides + wall A-H + flange E-F (3 bends)	QTY 1 FR + 1 FL = 2 per vehicle	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 1/3	

FR-10-2 · Part 2 "strip" — A-B · B-C · shelf · inner plate (3 bends)

10 ga 0.135" (3.42 mm) · HR steel sheet A1011 CS · flat blank 12.478" × 2.716" · ≈ 1.15 lb each · QTY 1 FR + 1 FL = 2 per vehicle

FLAT PATTERN (as laser-cut, marked side up)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	93.9°	UP (toward)	R=t 0.135	5	X = 2.337
B2	86.1°	UP (toward)	R=t 0.135	6	X = 4.736
B3	79.2°	DOWN (away)	R=t 0.135	4	X = 7.044

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø1.510	1.110	1.358	laser
H2	Ø1.510	5.809	1.358	laser

FILES

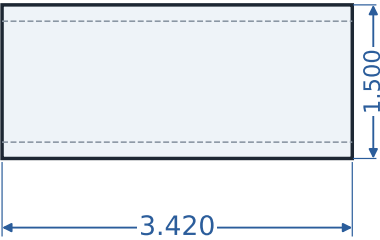
- FR-10-2_FR_cut.dxf (right, qty 1 / vehicle)
- FR-10-2_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-10-2_ref.dxf (reference: + bend lines, drill-later)

NOTES

- Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - Deburr all edges, remove dross.
 - Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.
- * X 79.2° opposite way; B 93.9°, C 86.1° up; corners 3/16"×45°

FR-10	Bracket: U-body + strip + tube	HR steel sheet A1011 CS 10 ga 0.135" (3.42 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FR-10-2 · Part 2 "strip" — A-B · B-C · shelf · inner plate (3 bends)		QTY 1 FR + 1 FL = 2 per vehicle	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 2/3

FR-10 · tube cut list



Tube OD 1.5 / ID 1.181 — cut 3.42" (through A-B and the shelf)

Stock: DOM round steel tube OD 1.500" × ID 1.181" (wall ≈ 0.160")

Cut length 3.420" ±1/32", square ends, deburr

QTY 1 per side × 2 (FR + FL) = 2 per vehicle

FR-10	Bracket: U-body + strip + tube	see stock above	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
Tube cut list		see quantities above	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 3/3

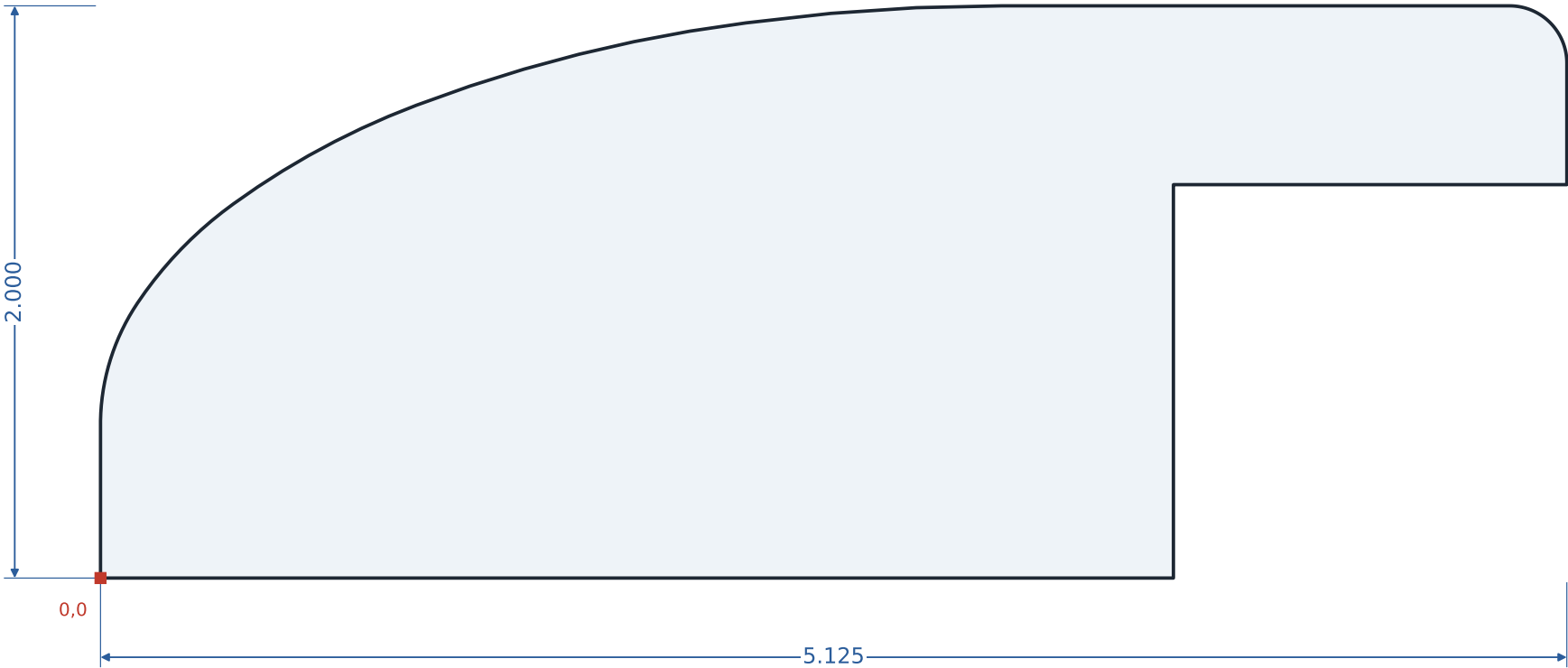
FR-11 · FR-11 plate (flat)

11 ga 0.120" (3.04 mm) · HR steel sheet A1011 CS · flat blank 5.125" × 2.000" · ≈ 0.25 lb each · QTY 2 FR + 2 FL = 4 per vehicle

FLAT PATTERN (as laser-cut, marked side up)

FILES

- FR-11_FR_cut.dxf (right, qty 2 / vehicle)
- FR-11_FL_cut.dxf (left = mirror, qty 2 / vehicle)



NOTES

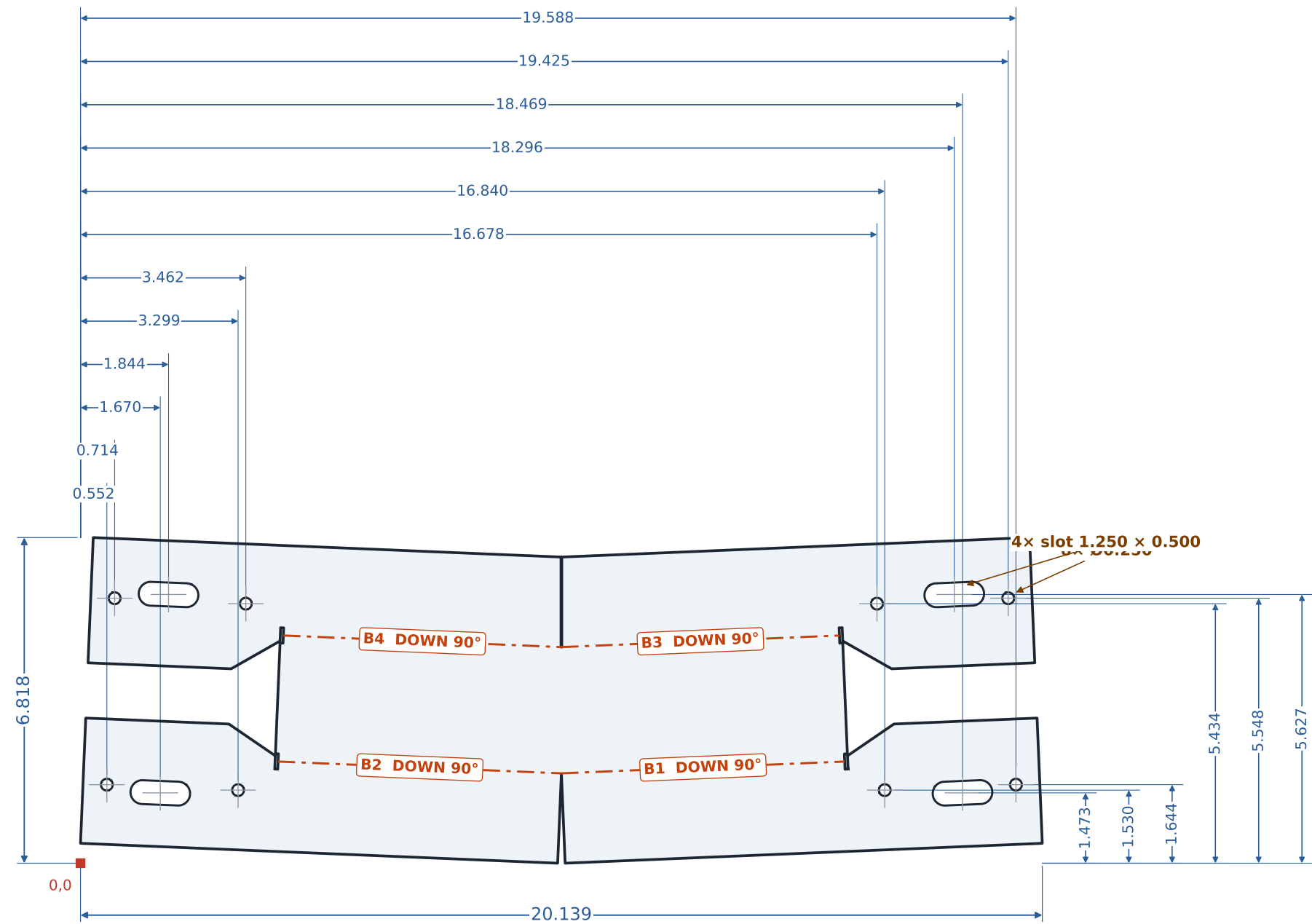
- 1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - 2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - 3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - 4. Deburr all edges, remove dross.
 - 5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.
- * flat, no bends; 2 per side, side by side; FL-11 = mirror

FR-11Flat plate	HR steel sheet A1011 CS 11 ga 0.120" (3.04 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FR-11 · FR-11 plate (flat)	QTY 2 FR + 2 FL = 4 per vehicle	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 1/1

FF-1 · Channel with ears — flat pattern, one piece (4 bends 90°, web on top)

10 ga 0.135" (3.42 mm) · HR steel sheet A1011 CS · flat blank 20.139" × 6.818" · ≈ 4.35 lb each · QTY 1 per vehicle

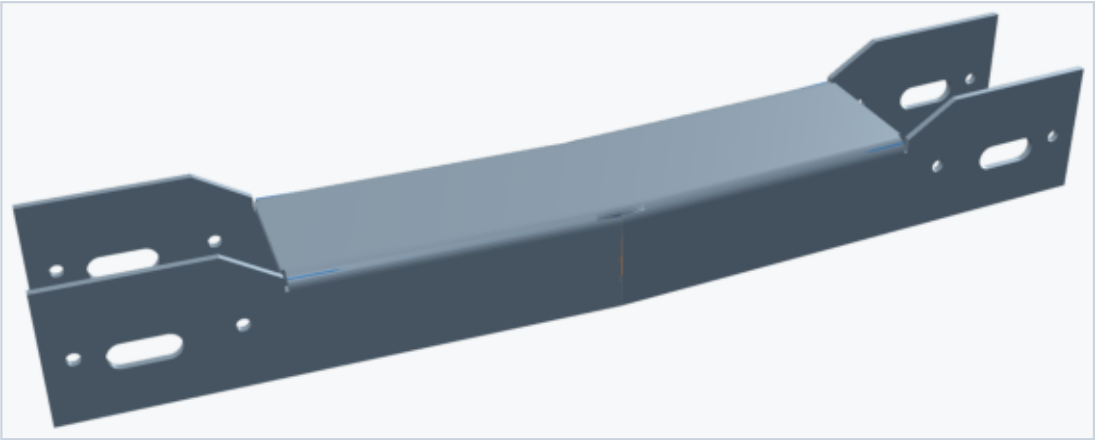
FLAT PATTERN (as laser-cut, marked side up)



NOTES

- 1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - 2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - 3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - 4. Deburr all edges, remove dross.
- * one blank: the web is a chevron 6"+6" (ends ¼" out); bend the walls 90° down on the dashed lines, each in 2 hits (6" to the middle); outer wall has a V-notch — closes after bending, weld it; inner wall has a slit — opens ~0.16" after bending, weld it; the ears reach past the bend line; 1/16" relief slit at the bend end

FORMED PART (after bending)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	DOWN (away)	R=t 0.135	1	(10.070, 1.881)→(16.002, 2.129)
B2	90.0°	DOWN (away)	R=t 0.135	2	(4.137, 2.129)→(10.070, 1.881)
B3	90.0°	DOWN (away)	R=t 0.135	3	(10.070, 4.525)→(15.892, 4.768)
B4	90.0°	DOWN (away)	R=t 0.135	4	(4.247, 4.768)→(10.070, 4.525)

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø0.250	19.588	1.644	laser
H2	Ø0.250	19.425	5.548	laser
H3	slot 1.250×0.500	18.469	1.473	laser
H4	slot 1.250×0.500	18.296	5.627	laser
H5	Ø0.250	16.840	1.530	laser
H6	Ø0.250	16.678	5.434	laser
H7	Ø0.250	3.462	5.434	laser
H8	Ø0.250	3.299	1.530	laser
H9	slot 1.250×0.500	1.844	5.627	laser
H10	slot 1.250×0.500	1.670	1.473	laser
H11	Ø0.250	0.714	5.548	laser
H12	Ø0.250	0.552	1.644	laser

FILES

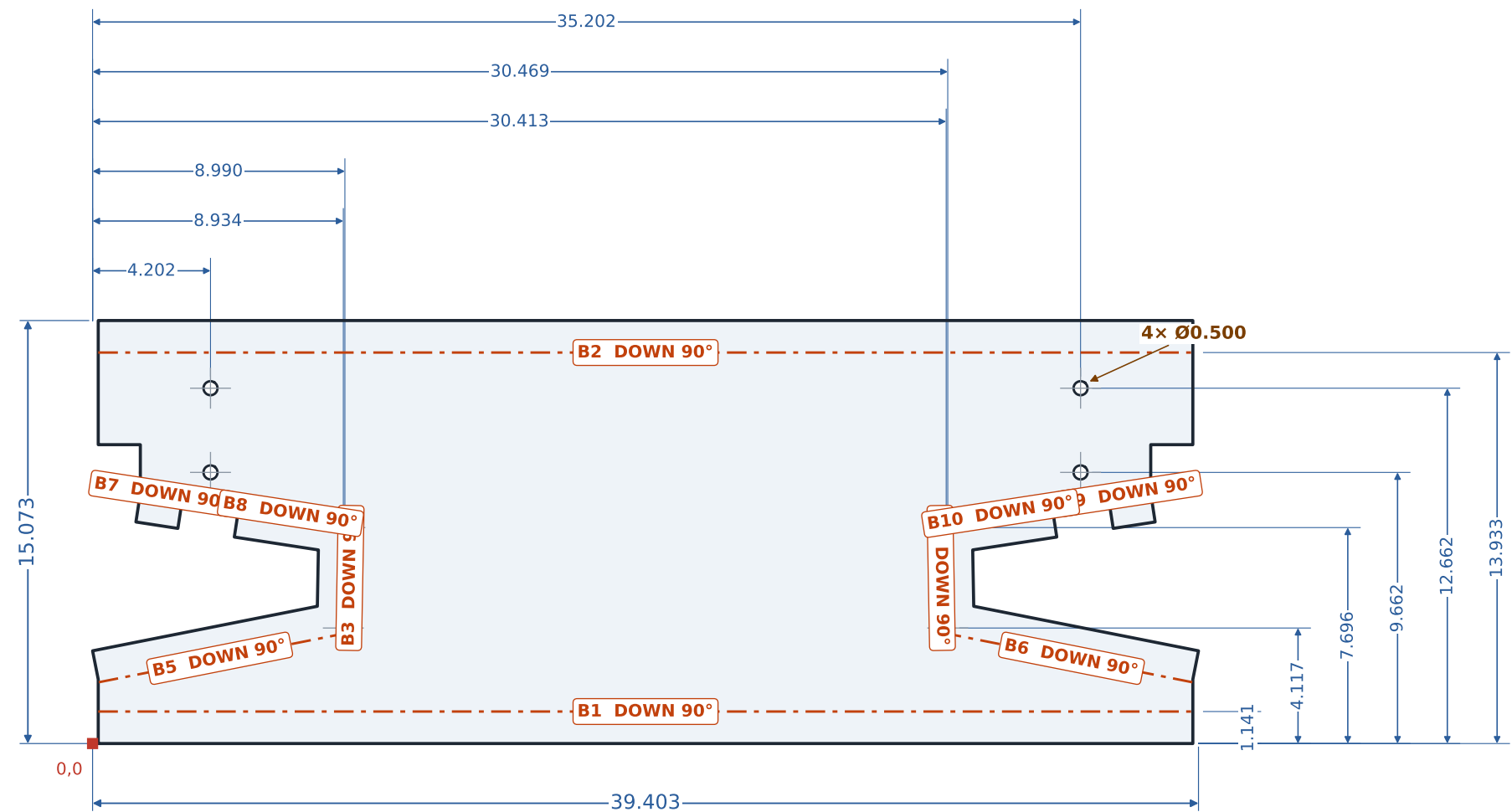
- FF-1_cut.dxf (qty 1 / vehicle)
- FF-1_ref.dxf (reference: + bend lines, drill-later)

FF-1	Front channel with ears (chevron 6"+6")	HR steel sheet A1011 CS 10 ga 0.135" (3.42 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FF-1 · Channel with ears — flat pattern, one piece (4 bends 90°, web on top)		QTY 1 per vehicle	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 1/1

FF-3-1 · FF3-1 top plate — flat pattern, 10 bends 90° down

11 ga 0.120" (3.04 mm) · HR steel sheet A1011 CS · flat blank 39.403" × 15.073" · ≈ 17.70 lb each · QTY 1 per vehicle

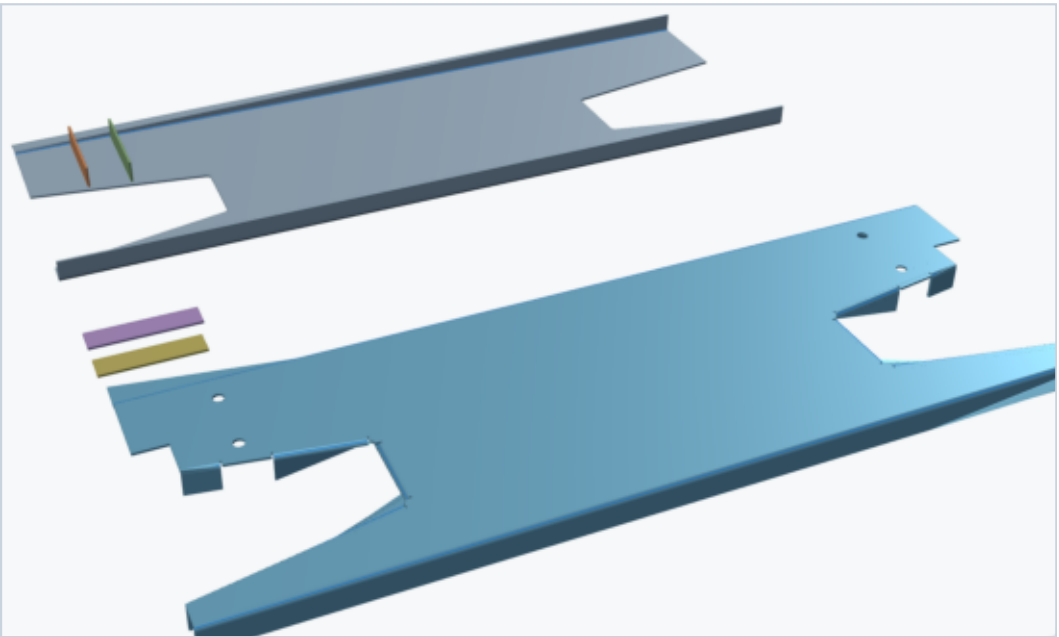
FLAT PATTERN (as laser-cut, marked side up)



NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 4. Deburr all edges, remove dross.
- * rims 1½" bent down on the long sides and around the whole notch; leg ends open; in the 4 inner notch corners the rims are mitred — a gap stays in the corner; 1/16" reliefs at the bend ends; the 4 Ø½ holes are cut right away

FORMED PART (after bending)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	DOWN (away)	R=t 0.120	3	Y = 1.141
B2	90.0°	DOWN (away)	R=t 0.120	4	Y = 13.933
B3	90.0°	DOWN (away)	R=t 0.120	5	(9.138, 4.256)→(9.197, 7.546)
B4	90.0°	DOWN (away)	R=t 0.120	6	(30.265, 4.256)→(30.206, 7.546)
B5	90.0°	DOWN (away)	R=t 0.120	7	(0.222, 2.185)→(8.833, 3.892)
B6	90.0°	DOWN (away)	R=t 0.120	8	(39.181, 2.185)→(30.570, 3.892)
B7	90.0°	DOWN (away)	R=t 0.120	9	(1.717, 9.022)→(3.212, 8.793)
B8	90.0°	DOWN (away)	R=t 0.120	10	(5.223, 8.486)→(8.878, 7.926)
B9	90.0°	DOWN (away)	R=t 0.120	11	(37.686, 9.022)→(36.191, 8.793)
B10	90.0°	DOWN (away)	R=t 0.120	12	(34.180, 8.486)→(30.526, 7.926)

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	cut-out 0.513×0.516	30.326	7.802	laser
H2	Ø0.500	35.202	12.662	laser
H3	Ø0.500	35.202	9.662	laser
H4	cut-out 0.514×0.504	30.387	4.011	laser
H5	cut-out 0.513×0.516	9.077	7.802	laser
H6	cut-out 0.514×0.504	9.016	4.011	laser
H7	Ø0.500	4.202	12.662	laser
H8	Ø0.500	4.202	9.662	laser

FILES

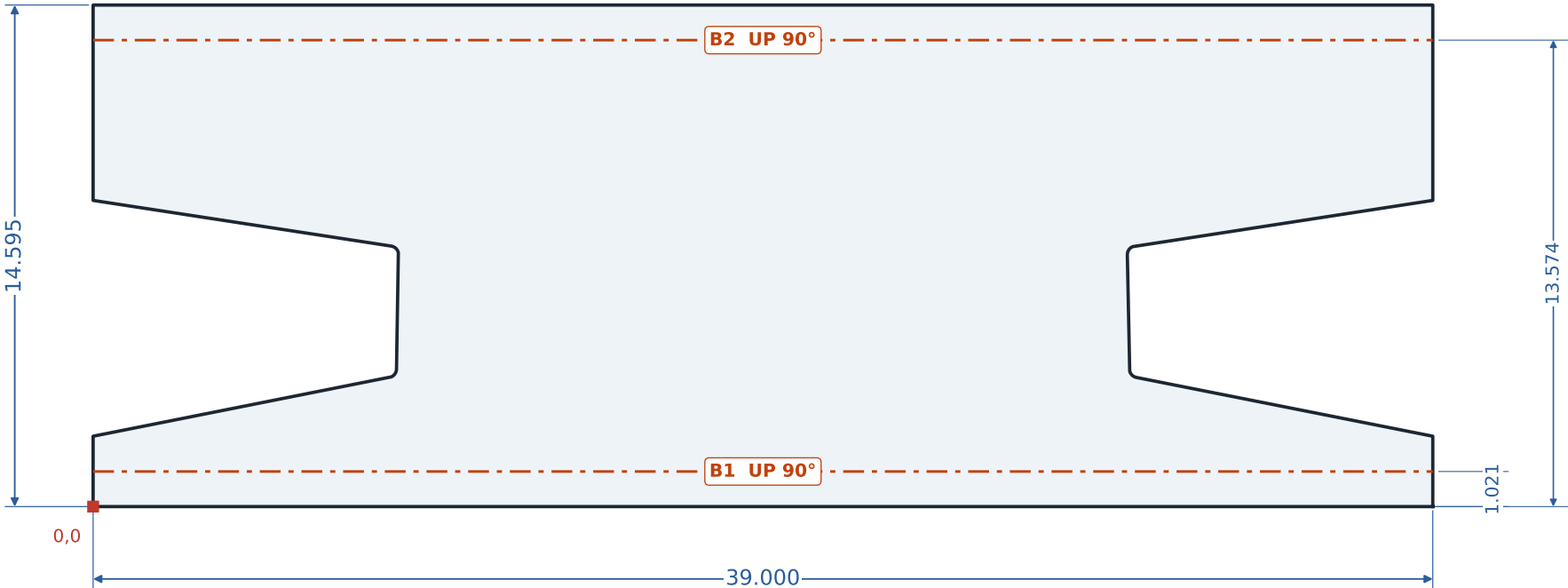
- FF-3-1_cut.dxf (qty 1 / vehicle)
- FF-3-1_ref.dxf (reference: + bend lines, drill-later)

FF-3	Cross box: top plate + lower plate + 4 passage walls	HR steel sheet A1011 CS 11 ga 0.120" (3.04 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FF-3-1 · FF3-1 top plate — flat pattern, 10 bends 90° down		QTY 1 per vehicle	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 1/4

FF-3-2 · FF3-2 lower plate — flat pattern, 2 bends 90° up

11 ga 0.120” (3.04 mm) · HR steel sheet A1011 CS · flat blank 39.000” × 14.595” · ≈ 16.12 lb each · QTY 1 per vehicle

FLAT PATTERN (as laser-cut, marked side up)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	UP (toward)	R=t 0.120	1	Y = 1.021
B2	90.0°	UP (toward)	R=t 0.120	2	Y = 13.574

FILES

- FF-3-2_cut.dxf (qty 1 / vehicle)
- FF-3-2_ref.dxf (reference: + bend lines, drill-later)

NOTES

- Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - Deburr all edges, remove dross.
- * rims on the long sides bent up 1" (to the FF3-1 underside), set in by t inside the FF3-1 rims; 1/16" reliefs at the bend ends

FF-3	Cross box: top plate + lower plate + 4 passage walls	HR steel sheet A1011 CS 11 ga 0.120” (3.04 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FF-3-2 · FF3-2 lower plate — flat pattern, 2 bends 90° up	QTY 1 per vehicle	Tol.: cut ±0.010” · bend ±1° · IR = t	2026-10-03	SHEET 2/4	

FF-3-3 · Passage wall 1⅛ × 4.69" (instead of the tube)

11 ga 0.120" (3.04 mm) · HR steel sheet A1011 CS · flat blank 4.687" × 1.125" · ≈ 0.18 lb each · QTY 2 per vehicle

FLAT PATTERN (as laser-cut, marked side up)

FILES

- FF-3-3_cut.dxf (qty 2 / vehicle)



NOTES

- 1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - 2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - 3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - 4. Deburr all edges, remove dross.
- * stands on FF3-2 under FF3-1, on the old tube wall line; weld top and bottom

FF-3	Cross box: top plate + lower plate + 4 passage walls	HR steel sheet A1011 CS 11 ga 0.120" (3.04 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FF-3-3 · Passage wall 1⅛ × 4.69" (instead of the tube)		QTY 2 per vehicle	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 3/4

FF-3-4 · Passage wall 1⅞ × 5.01" (instead of the tube)

11 ga 0.120" (3.04 mm) · HR steel sheet A1011 CS · flat blank 5.013" × 1.125" · ≈ 0.19 lb each · QTY 2 per vehicle

FLAT PATTERN (as laser-cut, marked side up)

FILES

- FF-3-4_cut.dxf (qty 2 / vehicle)



NOTES

- 1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - 2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - 3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - 4. Deburr all edges, remove dross.
- * stands on FF3-2 under FF3-1, on the old tube wall line; weld top and bottom

FF-3	Cross box: top plate + lower plate + 4 passage walls	HR steel sheet A1011 CS 11 ga 0.120" (3.04 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FF-3-4 · Passage wall 1⅞ × 5.01" (instead of the tube)		QTY 2 per vehicle	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 4/4

FF-4 · tube cut list



Square tube 1 ½" — cut 39"

Stock: Square steel tube 1 ½" × 1 ½" × 3/16" wall (A500)

Cut length 39.000" ±1/32", square ends, deburr

QTY 1 per vehicle

wall 3/16"

FF-4	Square tube 1 ½"	see stock above	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
Tube cut list		see quantities above	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 1/1